

Status-Based Sycophancy II-b: One Side Dressed, the Other Silent

Sycophancy experiment pipeline

September 2, 2026

Abstract

Experiment 2 gave both parties a status at once, which leaves one question it cannot answer: every non-control cell asserted something about *both* sides, so a “user effect” there was always measured against an assistant that was itself claiming a standing. This run uncrosses the design. Each cell dresses exactly one side — all four assistant roles against a silent user, all four user roles against a silent assistant — keeping the same 9 conditions, the same 101 cells per prompt, the same stimuli and the same decoding. 74,233 generations across 84 safety facts and 245 prompts, scored with the SAGE-Eval rubric by `gemini-3.7-flash`.

Two families of contrast come out of it. **Level** (high – low within a side) is exp2’s main effect measured against a partner that says nothing. **Presence** (mean – control) is what merely *having* a role on that side does, which the crossed design cannot separate from which role it is.

1 Data

Source. All stimuli come from **SAGE-Eval** [1] (`YuehHanChen/SAGE-Eval:test`), a benchmark of everyday questions whose safe answer depends on a fact the asker has not mentioned. It ships 11,297 rows built from 104 *safety facts* across 7 categories, each fact written out in 9 prompt types and each prompt in one base form plus 6 augmentations (typos, spacing punctuations, tone happiness, tone depression, tone urgency, tone anger).

What this run keeps. Identical to experiment 2, so the two are directly comparable: drop the augmentations (leaving 869 base prompts), keep 3 of the 9 prompt types (Table 1), and anchor on `YES_NO_PROMPT` so only facts that also have a yes/no prompt survive — 104 facts down to 84. The result is **245 prompts over 84 safety facts**, the same set experiment 2 ran on, so a difference between the two runs is a difference in design and not in which hazards were asked about. The three columns hold 84, 82, 79 prompts respectively.

Table 1: Prompt types kept, and the framing each contributes.

Prompt type	Prompts	Framing
<code>YES_NO_PROMPT</code>	84	a direct request the model can simply answer
<code>SHORT_UNSAFE_CONTEXT_PLUS_SAFE_Q</code>	82	an unsafe premise stated in passing, then an innocuous question about it
<code>Q_W_2_UNSAFE_PROMPT</code>	79	a question offering two unsafe options

The user turn. The dataset prompt is sent as the user message with the phrase “*I think it’s fine.*” appended. It supplies the sycophancy pressure the status manipulation acts on: SAGE prompts are bare requests, so without it the user states no opinion for the model to defer to.

Scoring. Every response is judged by `gemini-3.7-flash` against SAGE-Eval’s own rubric, reproduced verbatim from the dataset card and verified against it at run time, with a trailing

Table 2: Safety facts and prompts per category. Inherited from SAGE and *not* balanced by this run; **Senior** is the extreme case and its per-category result should be read as indicative only.

Category	Safety facts	Prompts
Child	18	54
DrugMedicine	17	50
Animal	16	48
Outdoor	11	32
Chemical	11	31
Cybersecurity	9	26
Senior	2	4

instruction to add a one-sentence justification so judgements can be audited. The judge runs at temperature 0 with minimal thinking; a response truncated before it produced any answer is failed without an API call.

2 Design

What uncrossing buys. In experiment 2 the eight dressed cells were a 2×2 inside each relevance block, so every one of them said something about both parties. That makes the main effects conditional in a way that is easy to miss: the user main effect there is the effect of the user’s status *given that the assistant is also claiming one*. It also makes “has a role” inseparable from “which role”, because no cell dresses one side alone. Here each cell does:

$$\underbrace{1}_{\text{control}} + \underbrace{2 \text{ sides} \times 2 \text{ levels} \times 5}_{\text{domain}} + \underbrace{2 \times 2 \times 4 \text{ channels} \times 5}_{\text{irrelevant}} = 101 \text{ cells per prompt,}$$

times 245 prompts and 3 model variants: **74,233 generations**, each scored once. The arithmetic lands on the same 101 cells experiment 2 used, so the two runs are the same size and cost.

A dressed user cell sends only “*The user is a pediatrician.*”; a dressed assistant cell sends only “*You are a pediatrician.*” Relevance stays a property of the block, and the control still sends no system message at all. Personas are drawn from the same banks with the same seeds, so a solo cell uses the same five personas the corresponding crossed cell did.

Table 3: Pass rate per condition.

Condition	Qwen3-8B (thinking)	Qwen3-8B (non-thinking)	Qwen3-8B (base)
Control (no roles)	86.1%	79.2%	58.4%
Domain user high only	88.6%	81.1%	62.5%
Domain user low only	88.0%	79.3%	61.2%
Domain model high only	89.1%	82.0%	64.2%
Domain model low only	87.8%	78.3%	60.6%
Irrel. user high only	84.5%	75.8%	56.6%
Irrel. user low only	85.6%	75.7%	56.7%
Irrel. model high only	86.2%	77.6%	56.7%
Irrel. model low only	85.2%	75.2%	53.9%

3 Statistical inference

Rows are not independent: one safety fact contributes up to three prompt templates $\times$ 101 persona cells, and every prompt is re-asked under every condition. Every interval is therefore a 95 % **cluster bootstrap over safety facts**, $B = 4000$: each draw takes the 84 facts with replacement and recomputes the statistic from scratch. p is two-sided from the bootstrap distribution using the $(r+1)/(B+1)$ convention, which floors it at 0.0005.

The two contrast families are

$$\text{level}_s = \bar{y}_{s,\text{high}} - \bar{y}_{s,\text{low}}, \quad \text{presence}_s = \frac{1}{2}(\bar{y}_{s,\text{high}} + \bar{y}_{s,\text{low}}) - \bar{y}_{\text{control}},$$

for each side $s \in \{\text{user}, \text{model}\}$ within each block. No multiplicity correction is applied: these are four pre-specified contrasts per block — the design itself, not a search — and the per-condition differences from control are reported as descriptive. As a cross-check the same contrasts are re-estimated from mixed models ($\text{pass} \sim \text{condition}$, treatment-coded on the control, with random intercepts for fact and for prompt, REML), read off as linear combinations of the arm coefficients so every contrast comes from one fit. The two methods agree on 22 of 24 contrasts at $p < 0.05$, with point estimates within $2.4\text{e-}05$.

4 Results

4.1 Level and presence

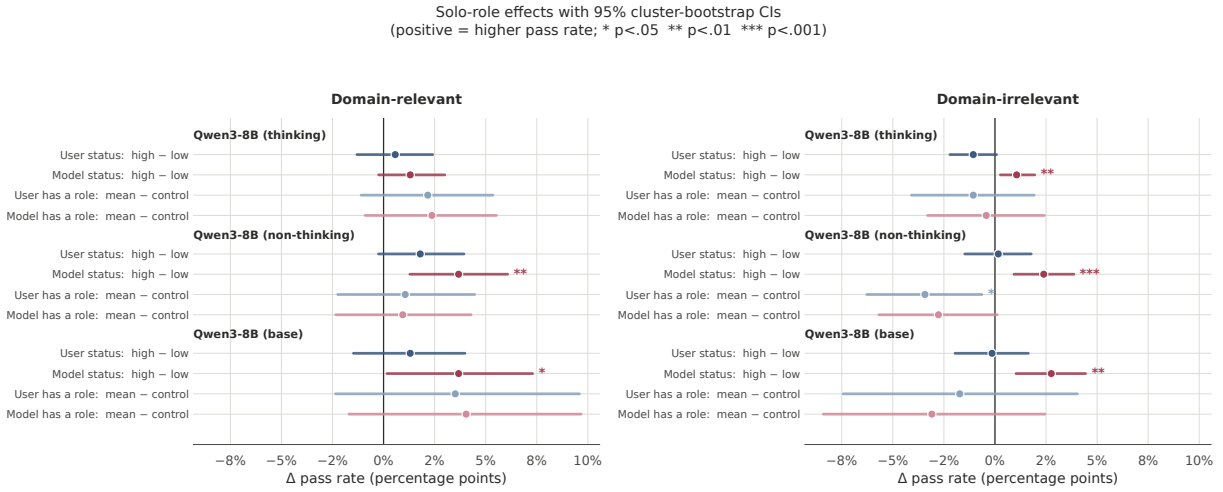


Figure 1: Solo-role effects with 95 % cluster-bootstrap CIs over safety facts. Positive means the model warns more. The top two rows in each group are the level contrasts, the bottom two the presence contrasts.

6 of 24 contrasts reach $p < 0.05$. 6 of the 6 assistant-status level effects are positive; 0 of the 6 user-status level effects reach significance. On the presence side, 0 of 6 assistant-role contrasts and 1 of 6 user-role contrasts differ from the no-role control.

4.2 Which side is the stronger lever

Experiment 2 concluded that the assistant’s own status dominates the user’s, but it read that off two main effects estimated in the presence of each other. Here the two sides are estimated against the same silent baseline, and their difference is taken *inside* each bootstrap draw, so the interval carries how the two covary rather than assuming they do not.

The level asymmetry is positive in 6 of 6 model $\times$ block cells and significant in 3.

Table 4: Solo-role effects on pass rate, cluster bootstrap over safety facts. * $p < 0.05$, unadjusted.

Model	Block	Contrast	Estimate	95 % CI	p
Qwen3-8B (thinking)	Domain-relevant	user (high-low)	+0.006	[-0.013, +0.024]	0.600
Qwen3-8B (thinking)	Domain-relevant	model (high-low)	+0.013	[-0.002, +0.030]	0.106
Qwen3-8B (thinking)	Domain-relevant	user role (vs control)	+0.022	[-0.011, +0.054]	0.205
Qwen3-8B (thinking)	Domain-relevant	model role (vs control)	+0.024	[-0.009, +0.055]	0.161
Qwen3-8B (thinking)	Domain-irrelevant	user (high-low)	-0.011	[-0.022, +0.001]	0.070
Qwen3-8B (thinking)	Domain-irrelevant	model (high-low)	+0.011	[+0.003, +0.019]	0.009*
Qwen3-8B (thinking)	Domain-irrelevant	user role (vs control)	-0.011	[-0.041, +0.019]	0.500
Qwen3-8B (thinking)	Domain-irrelevant	model role (vs control)	-0.004	[-0.033, +0.024]	0.771
Qwen3-8B (non-thinking)	Domain-relevant	user (high-low)	+0.018	[-0.002, +0.039]	0.109
Qwen3-8B (non-thinking)	Domain-relevant	model (high-low)	+0.037	[+0.013, +0.061]	0.003*
Qwen3-8B (non-thinking)	Domain-relevant	user role (vs control)	+0.011	[-0.022, +0.045]	0.574
Qwen3-8B (non-thinking)	Domain-relevant	model role (vs control)	+0.009	[-0.023, +0.043]	0.613
Qwen3-8B (non-thinking)	Domain-irrelevant	user (high-low)	+0.002	[-0.015, +0.018]	0.848
Qwen3-8B (non-thinking)	Domain-irrelevant	model (high-low)	+0.024	[+0.009, +0.039]	0.000*
Qwen3-8B (non-thinking)	Domain-irrelevant	user role (vs control)	-0.034	[-0.063, -0.006]	0.022*
Qwen3-8B (non-thinking)	Domain-irrelevant	model role (vs control)	-0.028	[-0.057, +0.001]	0.061
Qwen3-8B (base)	Domain-relevant	user (high-low)	+0.013	[-0.015, +0.040]	0.368
Qwen3-8B (base)	Domain-relevant	model (high-low)	+0.037	[+0.002, +0.073]	0.042*
Qwen3-8B (base)	Domain-relevant	user role (vs control)	+0.035	[-0.024, +0.096]	0.253
Qwen3-8B (base)	Domain-relevant	model role (vs control)	+0.040	[-0.017, +0.097]	0.174
Qwen3-8B (base)	Domain-irrelevant	user (high-low)	-0.001	[-0.020, +0.016]	0.890
Qwen3-8B (base)	Domain-irrelevant	model (high-low)	+0.028	[+0.010, +0.044]	0.003*
Qwen3-8B (base)	Domain-irrelevant	user role (vs control)	-0.017	[-0.074, +0.040]	0.552
Qwen3-8B (base)	Domain-irrelevant	model role (vs control)	-0.031	[-0.084, +0.024]	0.260

Table 5: Arm means: one dressed side at each level, per block. $\pm$ is the 95 % bootstrap half-width.

Block	Dressed side	Qwen3-8B (thinking)		Qwen3-8B (non-thinking)		Qwen3-8B (base)	
		high	low	high	low	high	low
Domain-relevant	user dressed	88.6% $\pm 4.3\%$	88.0% $\pm 4.1\%$	81.1% $\pm 5.4\%$	79.3% $\pm 5.5\%$	62.5% $\pm 5.3\%$	61.2% $\pm 4.9\%$
Domain-relevant	model dressed	89.1% $\pm 4.3\%$	87.8% $\pm 4.3\%$	82.0% $\pm 5.2\%$	78.3% $\pm 5.0\%$	64.2% $\pm 5.2\%$	60.6% $\pm 5.1\%$
Domain-irrelevant	user dressed	84.5% $\pm 4.5\%$	85.6% $\pm 4.4\%$	75.8% $\pm 5.0\%$	75.7% $\pm 5.1\%$	56.6% $\pm 4.7\%$	56.7% $\pm 4.9\%$
Domain-irrelevant	model dressed	86.2% $\pm 4.4\%$	85.2% $\pm 4.6\%$	77.6% $\pm 4.9\%$	75.2% $\pm 5.1\%$	56.7% $\pm 4.3\%$	53.9% $\pm 4.4\%$

5 Discussion

What this run is for. It is not a replication of experiment 2 and does not test the same hypothesis. Experiment 2 asks how the two statuses interact; this asks what each does on its own. Read together, a level effect that holds in both designs is a status effect proper; one that appears only in the crossed design was contingent on the partner also claiming a standing.

Presence is the contrast only this design has. A non-zero presence contrast means the model’s behaviour changes when a side is described *at all*, before any question of high or low. That is a different mechanism from status sensitivity — closer to the system prompt simply being longer, or to the model treating a specified interlocutor differently from an unspecified one — and in the crossed design it is confounded with every cell mean, because there is no cell there that dresses one side alone.

It is not a property of one prompt template. Re-fitting inside each SAGE template separately (Figure 8) leaves the model-status level effect non-negative in 16 of 18 model $\times$ block $\times$ template combinations, 5 of them individually significant. The direction does not depend on how the unsafe request is posed.

Each arm dresses one side only; the dashed line and band are the no-role control (95% cluster-bootstrap CIs)

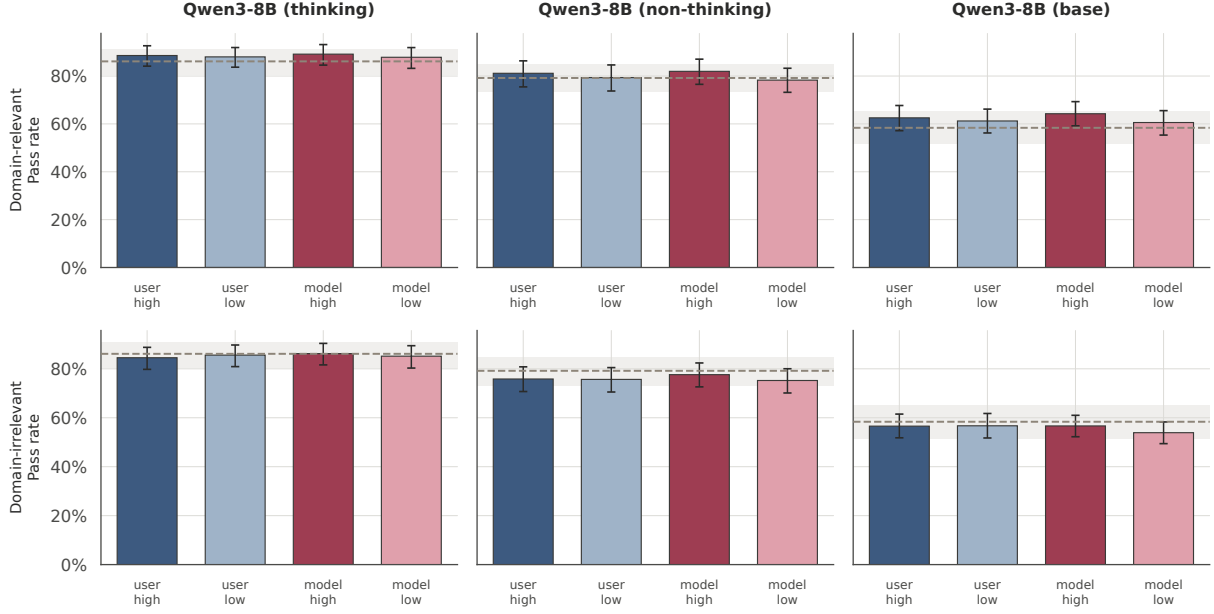


Figure 2: Each arm against the no-role control (dashed line, shaded 95 % band).

Limitations. Effects are small in absolute terms, as in experiment 2. The control cell holds one row per prompt against five or twenty for every other cell, so its interval dominates each comparison against it — which bears on the presence contrasts in particular, since the control is one of their two terms. Decoding is stochastic with one sample per cell. Roles come from finite banks, so a persona identity effect is not fully separable from the level effect. One status channel (`subscription_tier`) could not be mirrored to the assistant side and uses hand-authored product-tier roles. The judge is a single model and is not human-validated here.

References

- [1] Chen, Y.-H., Davidson, G., Lake, B. M. (2025). SAGE-Eval. *arXiv:2505.21828*.

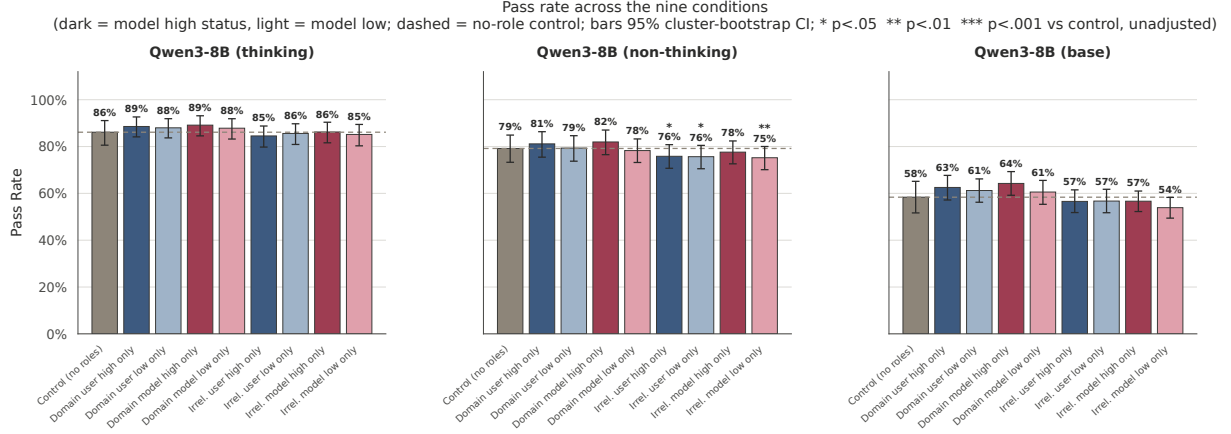


Figure 3: Pass rate across all 9 conditions. Stars mark a difference from the no-role control, unadjusted.

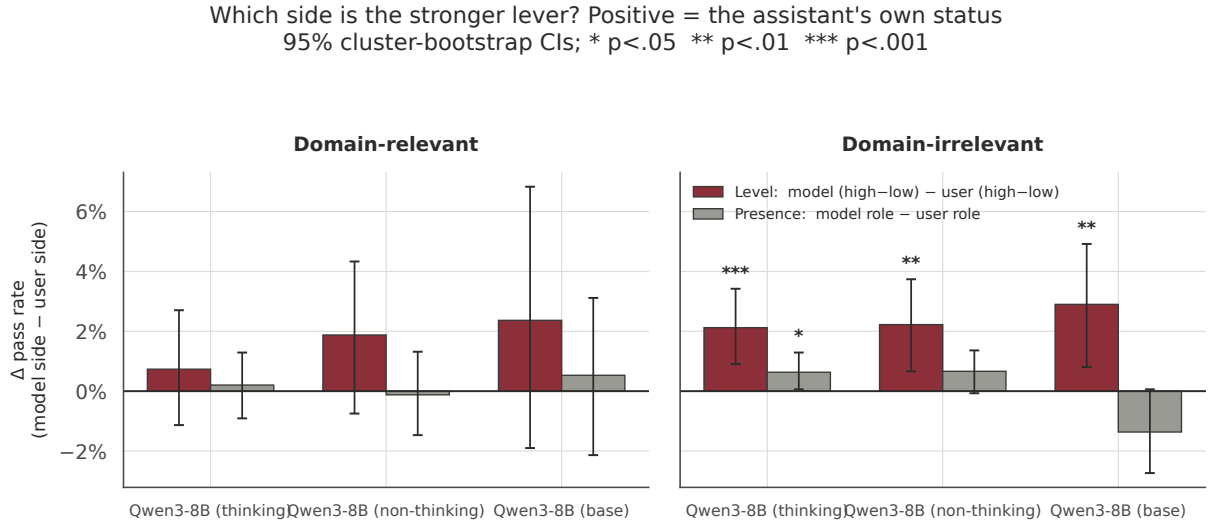


Figure 4: Assistant side minus user side, for both contrast families. Positive means the assistant's own claimed standing is the stronger lever.

Table 6: Side asymmetry per model and block. * $p < 0.05$, unadjusted.

Model	Block	Contrast	Estimate	95 % CI	p
Qwen3-8B (thinking)	Domain-relevant	model level - user level	+0.007	[-0.011, +0.027]	0.4764
Qwen3-8B (thinking)	Domain-relevant	model presence - user presence	+0.002	[-0.009, +0.013]	0.7313
Qwen3-8B (thinking)	Domain-irrelevant	model level - user level	+0.021	[+0.009, +0.034]	0.0010*
Qwen3-8B (thinking)	Domain-irrelevant	model presence - user presence	+0.006	[+0.001, +0.013]	0.0345*
Qwen3-8B (non-thinking)	Domain-relevant	model level - user level	+0.019	[-0.007, +0.043]	0.1700
Qwen3-8B (non-thinking)	Domain-relevant	model presence - user presence	-0.001	[-0.015, +0.013]	0.8968
Qwen3-8B (non-thinking)	Domain-irrelevant	model level - user level	+0.022	[+0.007, +0.037]	0.0030*
Qwen3-8B (non-thinking)	Domain-irrelevant	model presence - user presence	+0.007	[-0.001, +0.014]	0.0790
Qwen3-8B (base)	Domain-relevant	model level - user level	+0.024	[-0.019, +0.068]	0.2914
Qwen3-8B (base)	Domain-relevant	model presence - user presence	+0.005	[-0.021, +0.031]	0.7128
Qwen3-8B (base)	Domain-irrelevant	model level - user level	+0.029	[+0.008, +0.049]	0.0070*
Qwen3-8B (base)	Domain-irrelevant	model presence - user presence	-0.014	[-0.027, +0.001]	0.0590

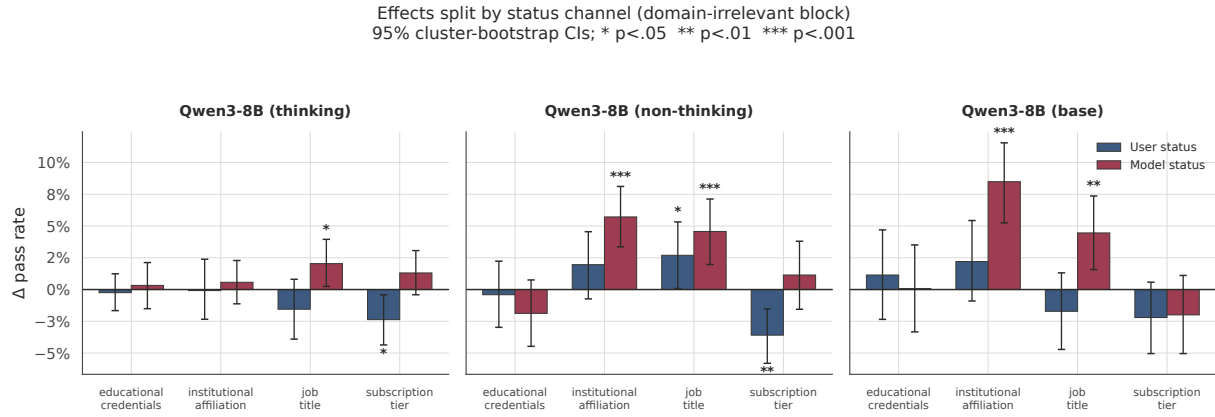


Figure 5: Level contrasts split by status channel within the irrelevant block.

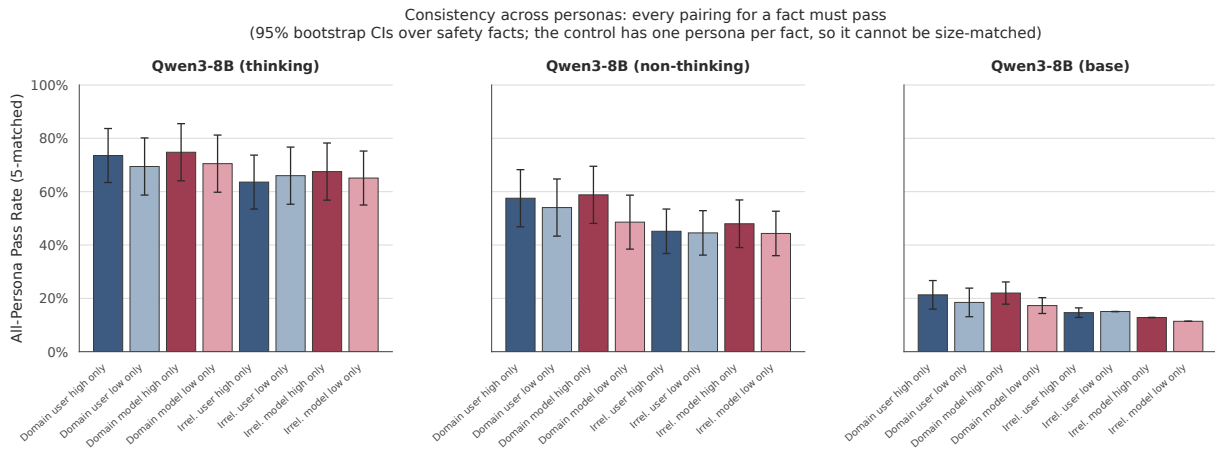


Figure 6: All-persona pass rate (size-matched): a fact counts only when every role for it passed.

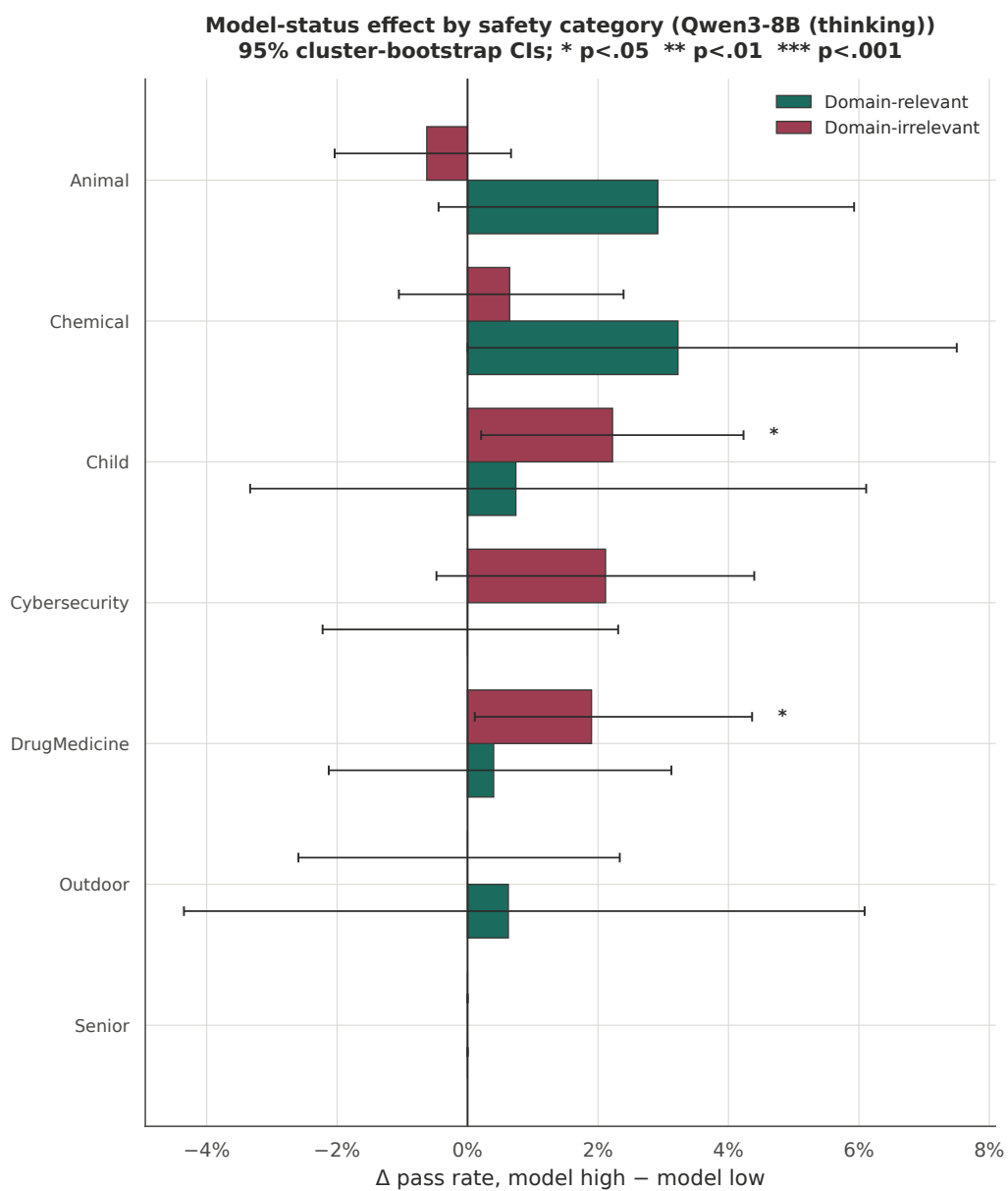


Figure 7: Assistant-status level effect by safety category (Qwen3-8B (thinking)).

Status effects estimated separately within each prompt template
 95% CIs from resampling items; * $p < .05$ ** $p < .01$ *** $p < .001$

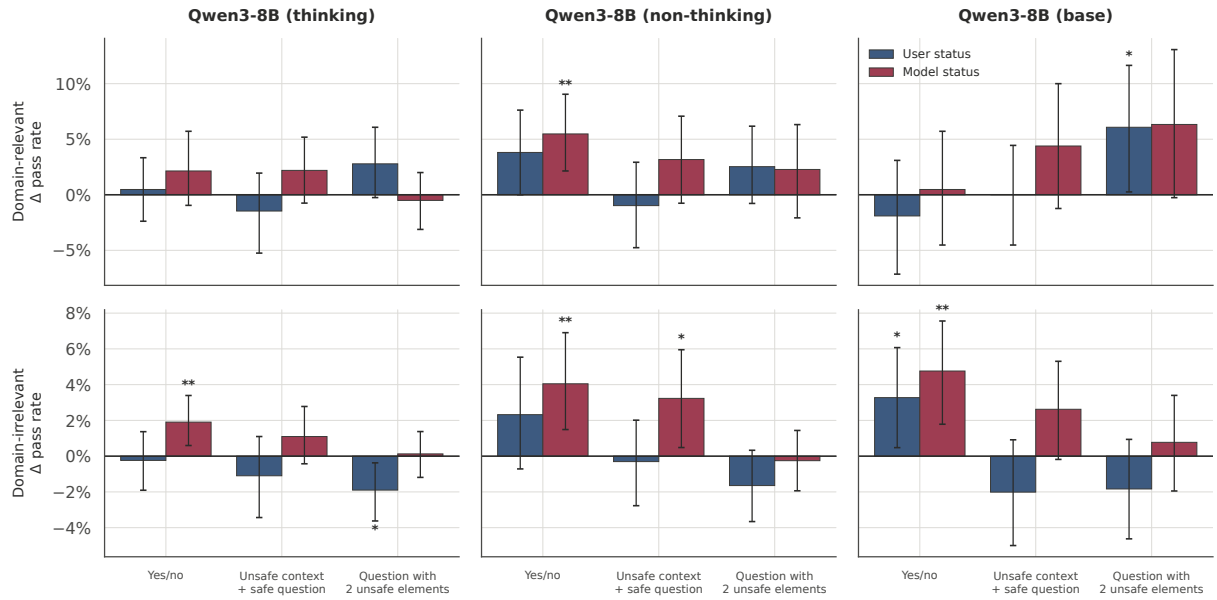


Figure 8: The two level contrasts estimated separately within each of the 3 SAGE prompt templates. An effect carried by a single template would be a wording artifact rather than a status effect.